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An ultrasonic feeding mechanism for continuous aerosol generation from cohesive powders

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ABSTRACT

The dispersion of dry, cohesive micro- and nanosized powders has wide applications ranging from medical and environmental science to manufacturing technology. Being able to disperse the dry powder at a stable concentration over periods of time (∼hour to several hours) is challenging to achieve, which we attempt to address with a novel dry powder dispersion technology. A portable, cost effective and simple dry powder dispersion device that uses ultrasonic energy for continuous feeding of aerosol particles to a de-agglomeration device is developed and tested with commercially available powders. The study is performed using 5 μm polyamide powder and rutile TiO_2 powders of 500, 100, and 30 nm mean size of primary particles. The ultrasonic dispersion device proved to be able to disperse all of these powders, which represent low-density material to high-density metal oxides, with stable concentration over periods that extend to an hour. From aerosol measurements to obtain the size distribution of various size particles, it is seen that the highly agglomerated powder particles are not easy to de-agglomerate as we go to the lower (nominal) sized powders because of the stronger inter-particle adhesion forces between nanoparticles compared to microparticles. The scanning electron microscopy (SEM) images of particles in their native powder state and those collected after the dispersion showed significant difference in the agglomeration of the powder particles, which suggests partial success of the turbulent jet de-agglomeration employed.

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Introduction

The dry dispersion of cohesive micro and nanopowders has applications in inhalation toxicology, powder processing, spread of pollutants, soil erosion and notably in a paradigm of additive manufacturing known as aerosol deposition method (Jun and Maxim 1999; Hanft et al. 2015), also known as vacuum cold spray (Fan et al. 2006; Liu et al. 2010), and vacuum kinetic spray (Cao et al. 2013; Kwon et al. 2014). Aerosol deposition method is a scalable, room temperature and solvent-free route for processing dense ceramic coatings in which sub-10 μm aerosol particles are impacted at speeds 300–1000 m/s on diverse substrates (metal/ceramic/glass/plastic) at reduced pressures (∼500 Pa). The mechanism of coating formation, termed as room temperature impact consolidation (RTIC), involves the conversion of the kinetic energy of impacting particles to compressive strain energy in the deposited particle fragments, subsequently compacting them into a dense layer for coating to build up at a high rate ∼ microns per minute (Akedo 2008). This method is particularly suitable for producing thin coatings of ceramic materials (Sayer and Sreenivas 1990). The adherence, thickness, porosity and surface area per unit mass of the produced thin film ceramics are highly sensitive to the morphology and extent of agglomeration of the particles used and their kinetic energy at impact. The design and quality control of industrial scale coating processes require strict control of the morphology and extent of agglomeration of impacting particles. For cost-effective manufacturing, it is favorable to source raw materials as dry powders produced in large quantities using currently well-established scalable methods that have a production rate of the order of kilograms per hour and high selectivity in the physical and chemical properties of the powders (Pratsinis 2010; Schimmoeller, Pratsinis, and Baiker 2011; Buesser and Pratsinis 2012). For diverse applications and basic scientific studies designed to

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understand the interplay of particle morphology, aggregation and kinetic energy on coating formation, it is necessary to generate nano- and micrometer sized aerosol particles at large concentrations ($>10{,}000\,cm^{-3}$) for extended periods of time (~few hours of continuous operation) from raw materials in dry powder form. In this article, we describe a dispersion mechanism that uses ultrasonic energy for dry dispersion and kinetic energy of a turbulent jet for de-agglomeration of cohesive powder particles for the continuous generation of µm- and nm-sized aerosols.

Dry dispersion of powders is often favored over wet dispersion (Biskos et al. 2008) and gas-to-particle conversion methods (Pratsinis 2010; Davide and Sankaran 2011) for its relative simplicity in process equipment, ability to preserve chemical purity and crystalline structure from the powder to aerosol phase, solvent-free and room-temperature operation. Powders are dispersed by injecting the constituent particles into a main gas flow that has sufficient kinetic energy to break them up into smaller particles and transport them as aerosols. However, this process presents itself with several challenges. Powders composed of sub-10 µm particles have high inter-particle and particle-wall adhesion energy. The adhesion forces consist of van der Waals, electrostatic and capillary forces (Beaudoin et al. 2015). It is well known from particle–particle and particle–surface adhesion models that the adhesion energy scales with inverse particle size to the 4th–6th power (Hiemenz and Rajagopalan 1997). This has important consequences for nanopowders that are highly cohesive and therefore very difficult to detach from the walls of process equipment during transport in tubing and while dispersion with moving parts such as blades, mixers or fans. Many prior efforts have looked into generating aerosols from dry powders using eductors, venturis, nozzles, mixers and fluidized beds as reviewed in detail elsewhere (Calvert, Ghadiri, and Tweedie 2009).

Aerosolization of dry powders can be viewed as a two-step process: The first step is the separation of the interfacial contact between powder particles and the surface on which they are resting (wall of process equipment or other particles). This requires that sufficient mechanical energy greater than the energy of adhesion for a given powder be supplied. The adhesion energy of powder particles follows a distribution that is affected by the material properties and the particle size distribution (Eichenlaub, Gelb, and Beaudoin 2004; Kumar et al. 2008; Jaiswal and Beaudoin 2011). For any quantum of mechanical energy applied to disperse a powder, particles with adhesion energy less than the applied energy will be dispersed while those with adhesion energy greater than the applied energy will not be dispersed. As time progresses, more and more of the easily dispersible particles are removed and mixed with the gas flow, leaving behind powder mass that is composed of highly adherent particles, thus making the sustained generation of particles at constant number concentration challenging. The second step in the aerosolization of dry powders is breaking up of the agglomerates of particles to form aerosols in the sub-micron size range from nanostructured, agglomerated powders. Again, the adhesion between primary particles of the powder scales with inverse size leading to high adhesion energy of the nanoagglomerates. The most convenient way to apply mechanical energy to break up the agglomerates is through turbulent kinetic energy of a gas and particle-wall collisions. Depending on the magnitude of the impact forces exerted on the particles, various degrees of de-agglomeration is likely to take place. Detaching powder particles down to individual particles is extremely difficult, as it requires a high amount of mechanical energy to overcome the inter-particle adhesion energy.

Recently, Tu et al. (2017), and before them, Tiwari, Fields, and Marr (2013) used a simple commercial vacuum generator to inject dry powders into a turbulent jet flow to promote de-agglomeration and dispersion following the original idea put forward by Tang et al. (2008). The extent of de-agglomeration is a function of the turbulence intensity in the vacuum generator used and has been shown to produce nm-sized aerosol. However, both Tang et al. (2008) and Tiwari, Fields, and Marr (2013) did not have a steady mechanism to feed the powder into the vacuum generator thereby leading to a burst or unsteady concentration output. Building on these works and to continuously feed powder into the vacuum generator, Tu et al. (2017) have designed a powder doser and pressure equilibrium unit to inject powder at a preset dosing rate. Using their novel design, they have demonstrated mode diameters in the 100–1000 nm size range of dispersed aerosol. However, the use of moving parts and the inherent nature of cohesive powders to stick to the walls would make the maintenance and sustained operation of their device presumably difficult. In contrast, To et al. (2009) have used the rapid expansion of powders in super-critical carbon dioxide to enable their de-agglomeration and have reported mode diameters in the nanosize range. The use of high pressures (several MPa) and liquid CO_2 can make this method potentially costly to operate. To

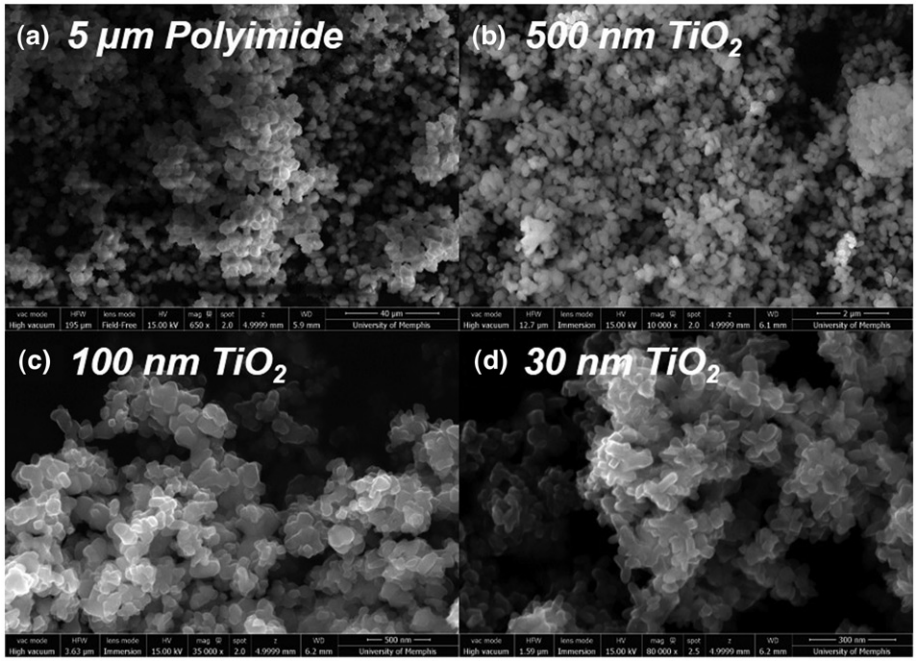


Figure 1. Scanning electron microscope images of powders used in the study as purchased. (a) 5 μm polyimide powder, (b) 500 nm TiO_2 powder, (c) 100 nm TiO_2 powder, and (d) 30 nm TiO_2 powder.

avoid these problems, following prior work on using acoustic energy to disperse dry powders (Weyel et al. 1984; Thorne 1994; Schmoll et al. 2009), we have used ultrasonic transducers to create a pressure field to acoustically levitate particles from a powder bed at ambient conditions. Schmoll et al. (2009) evaluated the dispersion of dry powders (21-nm TiO_2, 20-nm SiO_2 and SWCNTs) by the acoustic dry aerosol generator/elutriator (ADAGE) developed by Weyel et al. (1984). This device uses a loudspeaker to supply acoustic energy in the audible frequency range to a membrane on which desired powder is set. A stream of air carries the dispersed dust to form aerosol. Schmoll et al. (2009) concluded that the ADAGE system was able aerosolize SiO_2 and low-density SWCNT powders but was unable to aerosolize TiO_2 powders presumably due to the higher cohesiveness of TiO_2. In this work, ultrasonic pressure fluctuations are used to apply mechanical energy to lift particles off surfaces without moving parts. The interaction of a pressure field with particles has also been observed to promote agglomeration of them as well (Ng, Xiong, and Wan 2017). Motivated by these factors, our goal is to design a dry powder dispersion device that uses ultrasonic energy to liftoff particles from a powder bed at a near constant rate and can feed the particles to a turbulent flow de-agglomeration device. The de-agglomeration device (a commercial portable venturi pump) applies turbulent kinetic energy to break them up into smaller particles that was sized using standard aerosol instrumentation in the range of 20 nm–10 µm. The de-agglomerated aerosol particles were collected on scanning electron microscopy (SEM) substrates for image analysis to assess the dispersion and de-agglomeration. The measurements were aimed toward evaluating the dispersion system performance in terms of: (1) the ability to produce constant concentration of aerosol for long periods of time, (2) the size distribution of produced aerosol from dry TiO_2 powders, and (3) the size and morphology of particles collected from the aerosol phase compared with those from the powder phase to understand the extent of de-agglomeration. Finally, the choice of TiO_2 is motivated to demonstrate a device that is capable of aerosolizing cohesive powders of a dense material (bulk density of TiO_2 $\sim$ 4.23 g/cm^3 as several previous works have developed devices exclusively for low-density materials such as carbon nanotubes [Ma-Hock et al. 2009; Chen et al. 2012; Madl et al. 2012]). Along with high inter-particle cohesion forces, density also affects the ease of dispersion of powders. While low-density materials are easy to disperse, higher quantum of flow kinetic energy is required to lift and transport dense powders due to inertial settling in tubing and connectors. The choice of polyimide ($\sim$1.42 g/cm^3) and titanium dioxide ($\sim$4.23 g/cm^3) powders in this study is designed to demonstrate the potential of ultrasonic dispersion for a wide range of material densities of cohesive powders.

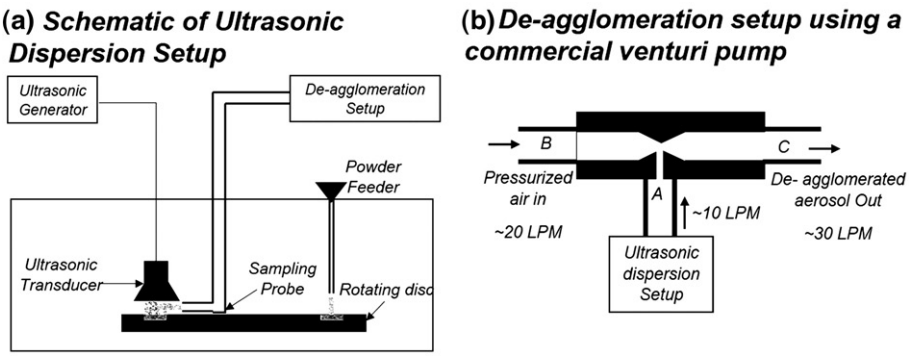


Figure 2. (a) Schematic representation of the ultrasonic dispersion setup constructed for this study. (b) Schematic of a commercial portable venturi pump used as an aspirator for sampling dispersed aerosol from the ultrasonic dispersion setup. A negative pressure is created at port A connected to the throat of the venturi due to the flow of particle-free air through the prefabricated nozzle in port B. Powder fragments from the ultrasonic dispersion setup are aspirated into the turbulent jet formed inside the venturi pump. The particles are well dispersed and thoroughly mixed due to the turbulence when they exit the venture pump through port C.

Experimental methods

Materials

The powders used in this study are 5 µm polyimide particles (Dantec Dynamics, #9080A3011) and rutile TiO_2 powders (US Research Nanomaterials Inc., #13463-67-7) of 500, 100 and 30 nm reported as the mean size of the primary particles by the manufacturer. The powders were used as purchased without any modification. The SEM images of the raw powders (Figure 1) show the non-spherical and highly agglomerated powder particles. Powders, especially titanium dioxide, absorb moisture from the air that makes them difficult to disperse in air as well de-agglomerate in a turbulent flow field. In all experimental runs, powders were heated to 100°C for 1 h to remove moisture and to ensure that the particles were readily levitated under the action of ultrasonic waves described shortly. The temperature of 100°C was chosen to remove moisture without causing melting or partial sintering of the particles (melting point of $TiO_2 \sim 1843$°C).

Ultrasonic dispersion setup

The ultrasonic dispersion setup depicted in Figure 2a consists of a slowly rotating disc (material AA 6061, 10 mm thick, radius 140 mm) with a groove (10 mm width and 2 mm depth) which is constantly replenished with the powder to be dispersed using a powder feeder system. The powder feeder system consists of a SS funnel constantly agitated by four small vibrating motors (Adafruit Industries LLC, Product ID 1201, 11,000 rpm, 5 V DC) attached to its outer surface and a metal mesh on the feeding end of the funnel to break up mm sized clumps of powder. An ultrasonic transducer (Stem Inc., SMBLTD63F25H2, 25 kHz) is suspended above the groove and the spacing between the disc and the transducer is manually adjusted. The gap distance (~0.5–2 cm) is tuned to form a standing wave pattern of pressure fluctuations that levitate the powder particles. The standing waves (manifested by the particle liftoff, see video available in the online supplementary files [SI]) appear at a precise gap distance at several locations above the disc. The location closest to the disc was chosen to minimize loss of power to the surroundings and maximize particle entrainment. The transducer is driven by an ultrasonic generator unit (Beijing Ultrasonic 1200 W) capable of varying the frequency (20–40 kHz) and power (0–1200 W). The generator was operated at maximum power of 1200 W and a frequency of 25.5 kHz, which was found to be the resonant frequency of the transducer. As the powder is agitated due to the pressure waves in the air column between the transducer and the rotating disc, it starts forming a small yet dense dust cloud of aerosol that is then sampled using a nominal $^1/_4''$ ID pipe for measurement. The powder particles that liftoff by ultrasonic excitation are broken down by the intense shear inside the de-agglomeration device. Figure S4 shows an SEM image of 500, 100 and 30 nm particles collected before entering the venturi pump and show that lumps of powders are seen to originate from the powder bed. These lumps, of the order of several microns up to 10 µm nominally, are presumably broken down inside the venturi pump. The aerosol exiting the venturi pump is introduced into a cyclone for removing particles larger than 10 µm prior to measurement and SEM sample collection. The removal of particles larger than 10 µm from the aerosol exiting the venturi pump is confirmed by the absence of particles larger than ~5 µm in all of the size distributions acquired for this study. The

negative pressure in the sampling probe is produced by connecting the same to the suction side of the venturi pump that acts as the de-agglomeration setup.

Turbulent de-agglomeration process

De-agglomeration is the breakdown of the powder particle clusters into smaller sizes. Two commercial venturi pumps (Airvac-TD260LSS, Airvac-UV143H) are used as vacuum generators, for creating negative pressure in the sampling probe to draw in the powder from the ultrasonic dispersion unit from port A (Figure 2b). The pressurized air introduced into port B of the venturi that consists of a prefabricated nozzle, attains sonic velocity and exits the nozzle as a turbulent jet. The intense shear forces due to turbulent fluctuations on the powder fragments sampled from the ultrasonic dispersion device breakdown the cluster forming smaller particles that exits through port C. All the measurements reported in this study correspond to a flow of 20.0 LPM of filtered, dry, particle-free air (@ 80 psig) through port B and a flow of 10.0 LPM sampled through port A, leading to a total volumetric flow (at atmospheric pressure and temperature) of approximately 30 LPM exiting through port C. This applies for both the venturi pumps used.

Details of the size distribution measurement using the TSI 3330 Optical Particle Sizer, *OPS* (0.3–10 µm) and standard differential mobility analysis (0.05–0.4 µm) is provided in SI Section S1. Also included are additional SEM images and pictures of the commercial venturi pumps Airvac-TD260LSS and Airvac-UV143H (Figure S2) used in this study as referenced appropriately in the main text.

Results and discussion

Stability of ultrasonic dispersion

Figure 3a shows the total concentration output from the dispersion and de-agglomeration system (Airvac TD260LSS venturi pump) during continuous operation. These measurements were obtained using the TSI 3330 OPS after removing particles larger than 10 µm and are integral of all the particles in the 0.3–10 µm range accessible to the OPS. Aerosols were generated for periods that lasted ~60 min and the concentration was monitored using the OPS for a 450 s time window. These monitoring windows were chosen randomly at the beginning, middle or end of the 60 min period to test the stability of the device. During these 450 s periods, the total concentration acquired every 15 s (each scan of the OPS lasted 15 s) is shown

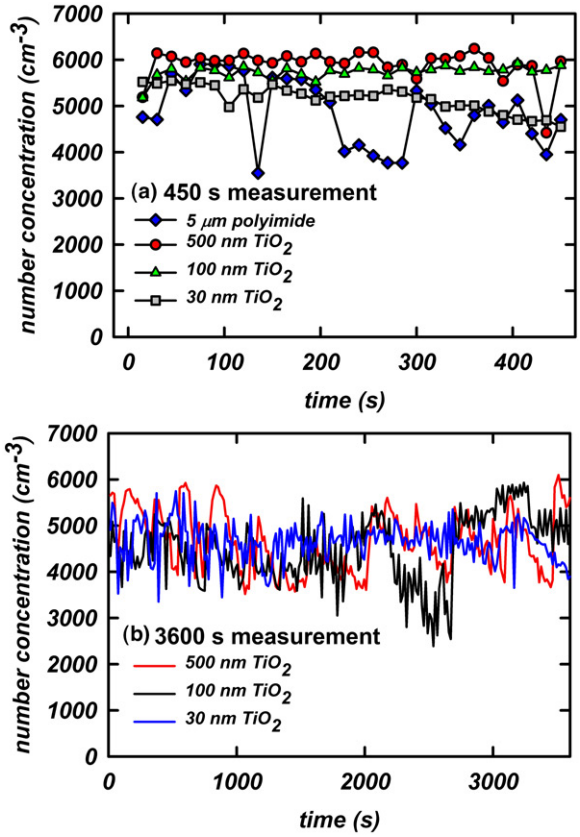


Figure 3. (a) Plots of number concentration of particles vs. time showing the stability of the ultrasonic dispersion device to generate particles from different powders using an Airvac TD260LSS venturi pump for de-agglomeration. The total concentration value plotted here are obtained from the Optical Particle Sizer (OPS) randomly selected during 450 s windows out of a measurement period that lasted 60 min. The reported concentration represents the particles in the 0.3–10 μm range. Each scan of the OPS was 15 s and scans were taken every 15 s during the measurement period. (b) Plots of number concentration of particles vs. time showing the stability using an Airvac UV143H venturi pump for de-agglomeration. Each scan of the OPS was 15 s and scans were taken every 15 s during the measurement period.

in Figure 3a. Alternately, in Figure 3b, concentration vs. time is shown for a 60 min period carried out using the Airvac-UV143H venturi pump (measurement runs distinct from the data shown in Figure 3a, acquired every 15 s similar to the data in Figure 3a) to show the stability of the dispersion mechanism for times up to 1 h for the three TiO_2 powders, with the exception of the 5 µm polyimide powder. The mean concentration with a 95% confidence interval and the standard deviation for the four powders studied is given in Table 1 for both the 450 s monitoring (Figure 3a) and 60 min monitoring (Figure 3b).

The confidence interval on the mean concentration shown in Figure 3a (~5.5% for the 5 µm powder and ≤2.2% for other powders) and low standard deviation (~14.8% for the 5 µm powder and ≤5.9% for other

Table 1. Summary of mean concentrations measured during 450 s (Figure 3a) and 60 min (Figure 3b) operation of the ultrasonic dispersion device.

	450 s monitoring (Figure 3a)			60 min monitoring (Figure 3b)		
Powder	Mean conc. (cm^{-3})	Std. dev. (%)	95% confidence interval on the mean	Mean conc. (cm^{-3})	Std. dev. (%)	95% confidence interval on the mean
5 μm polyimide	4869	14.8%	5.5%	–	–	–
500 nm TiO_2	5908	5.9%	2.2%	4674	14.1%	1.8%
100 nm TiO_2	5734	2.5%	0.1%	4498	16.1%	2.0%
30 nm TiO_2	5165	5.5%	2.0%	4651	8.9%	1.1%

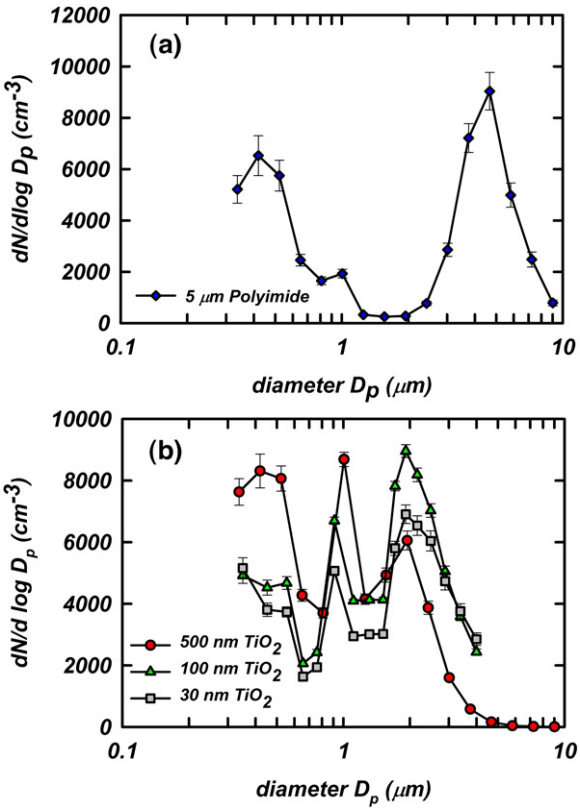


Figure 4. Size distribution of (a) 5 μm, (b) 500, 100, and 30 nm aerosol particles de-agglomerated using the TD260LSS venturi pump measured using the Optical Particle Sizer. The size distribution function as calculated by the Aerosol Instrument Manager software (Copyright: TSI) is reported in this paper. The error bars were calculated based on the statistical variation (95% confidence interval on the mean value) over 30 runs during data acquisition. Each scan of the OPS lasted 15 s and the measurement was carried out for 450 s in total.

powders) in reported concentrations show that the device is capable of maintaining steady concentration for short periods of 450 s. The 60 min measurements show slightly higher fluctuations in concentration as evidenced by the relatively high standard deviation of ~8.9–14.1% compared to the shorter time measurements. These fluctuations are attributed to the formation of powder lumps in the groove containing the feed powder. Over time, easily removable particles are dispersed from the powder bed leaving behind a population of relatively well-bonded particles. This phenomenon is commonly referred to as "caking of powders." To overcome this problem, we introduce ~1 mm stiff needles of steel in the circular groove to continuously rake up the powder (not shown in Figure 2a for simplicity). This greatly improved the stability of aerosol generation. In each case, it can be seen that the concentration is ~4000–6000 cm^{-3}, which is tunable and system specific. The output concentration can be potentially increased without affecting the flow field inside the vacuum pump and the turbulent de-agglomeration significantly by injecting a higher concentration of powder particles lifted from the powder bed using multiple transducers. However, further study will be required to design a scaled-up system for which the current study provides proof of concept. Further, the use of multiple transducers simultaneously can rapidly scale up the aerosol generation assuming that powder can be fed through a suitable hopper to ensure that there is powder in the groove at all times for the ultrasonic transducer to lift them and disperse them as aerosols. The concentration is possibly limited by the re-deposition rate of initial clusters that are lifted by ultrasonic energy, inertial losses in the transport tubing and physical space available to accommodate a certain number of transducers over the rotating powder bed. This mechanism provides a potential route for continuous aerosol generation necessary for dry particle deposition or spray based additive manufacturing as well as basic scientific investigations.

0.3 –10 μm size distribution

Figure 4 shows the size distribution function measured using the TSI 3330 OPS. These measurements were simultaneously obtained when the time series data shown in Figure 3a were collected. Figure 4a shows the size distribution of nominal 5 μm polyimide particles. As can be seen a distinct peak appears at ~5 μm. Analysis of SEM images (Figures 5a and S5) of particles collected from the gas phase confirm that the observed particles are nominally 5 μm, thereby confirming the aerosolization. We observed that the particles are found as individual entities in contrast to

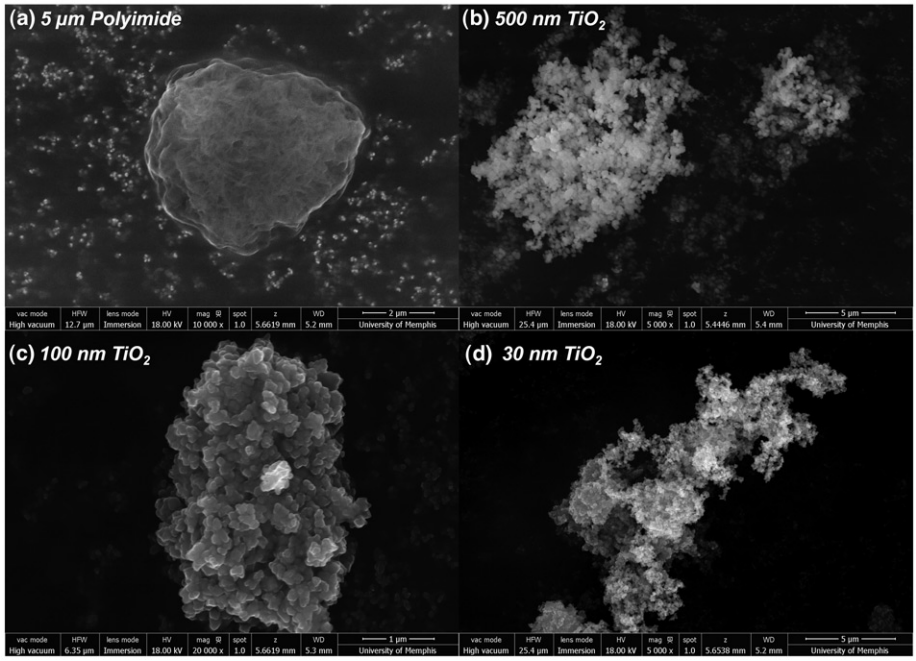


Figure 5. SEM images of particles collected from the gas phase to investigate the extent of de-agglomeration of the particles by the TD260LSS venturi pump.

the other sizes that will be discussed shortly. From considerations of inter-particle adhesion, 5 µm particles are expected to be de-agglomerated relatively easily and are used as visualization media for particle image velocimetry (PIV) methods in liquids and gases. Figure 4b shows the size distribution for 500, 100, and 30 nm powders. We find that all the three powders have a similar size distribution with peaks appearing at ~ 2, ~1 and ~0.5 µm. To understand the observed size distribution, we imaged these particles as collected from the gas phase, shown in Figures 5b–d. In sharp contrast to the 5 µm powder, the other three powders are found in a highly agglomerated state after "de-agglomeration" by the Airvac TD260LSS venturi pump. We attribute this to the relatively low level of turbulence intensity inside the TD260LSS that is unable to effectively break up the agglomerates of each type of particle. Here, it should be noted that the 500, 100 and 30 nm denote the primary particles reported by the manufacturer and bear no relation to the extent of agglomeration that the particles may undergo in powder or aerosol phase. Due to the dense nature of the agglomerates and to the relatively large sizes at which these were found in the SEM images (see Figure 5, and S6-1, S7-1, S8-1 in SI), we inferred that the OPS size distributions correspond to the ~µm scale agglomerates and not of nanoscale aerosol generated from the powder phase. We used energy dispersive X-ray spectroscopy (EDS) to evaluate the elemental composition of the collected particles. Figures S6-2, 3 show the EDS elemental map of Ti and O for 500 nm powders. Likewise, Figures S7-2 and S7-3 show the EDS maps for representative 100 nm particles and lastly Figures S8-2 and S8-3 show the same for 30 nm powder particles. In each case, it is seen that the particles sampled from the gas phase originated from our powder dispersion.

We elected to repeat the aerosol de-agglomeration with a venturi pump capable of generating higher levels of turbulence as quantified by its higher flow capacity and vacuum level generated. Further experiments were conducted with the Airvac UV143H vacuum generator and the size distribution measured using OPS is shown in Figure 6a. For the 500 nm powder, considerable particles are still seen at 2 µm but is reduced compared to the low intensity venturi (TD260LSS). For the 100 and 30 nm powders, a principle peak appears at ~0.5 µm although that size does not correspond to their true size. Compared to the size distributions in Figure 4, the peaks that formerly appeared at ~2 µm are now diminished and the intensity of the peaks at ~0.5 and ~1 µm have increased comparatively in Figure 6a for all the three TiO_2 powders. This shows that extent of de-agglomeration plays a key role is nanoscale aerosol generation from cohesive powders. If successfully de-agglomerated, due to the high concentration, re-agglomeration is also possible unless the generated aerosol is quickly diluted suitably. In this study, we did not observe any re-agglomeration because, such particles are expected to

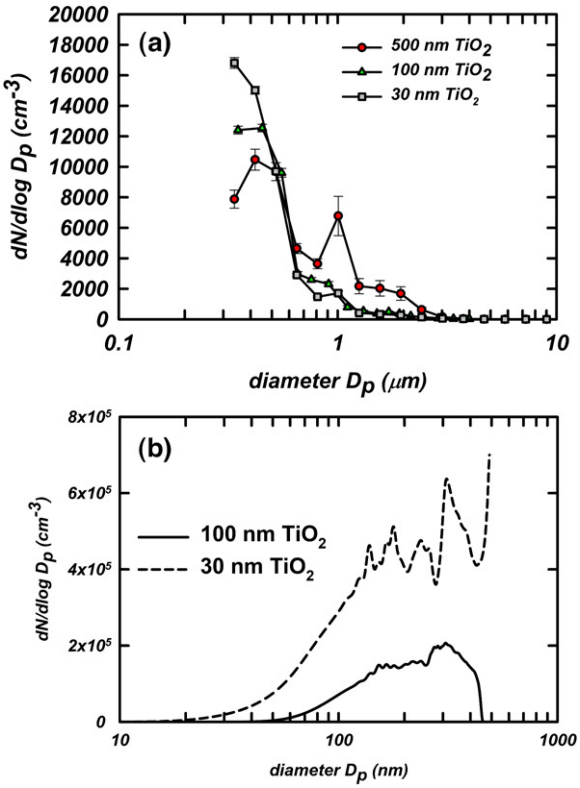


Figure 6. Size distribution of particles de-agglomerated using the UV-143H venturi pump using (a) OPS (The error bars were calculated based on the statistical variation (95% confidence interval on the mean value) 100 runs during data acquisition. Each scan of the OPS lasted 15 s and the measurement was carried out for 3600 s in total and (b) electrical mobility analysis. Error bars are not included for mobility measurements as they were of the order of the accuracy with which the flow rates were calibrated ~1% and hence are not shown on the plot.

be more open structures as opposed to the dense ~ μm particles that we observed (Figure 5). To understand nanoparticle generation using ultrasonic dispersion, size distributions were measured in the sub-0.4 μm range as described next.

Sub-0.4 μm size distribution

Differential mobility analysis was used to obtain the size distribution of particles in the sub-0.4 μm size range to assess nanoaerosol generation after de-agglomeration. The details of the measurement methods are given in Section S1, *SI*. Briefly, aerosol generated from powders was sampled from a dispersion bag (~1000 L) and charged using a soft X-ray bipolar charger (TSI 3088) prior to measurement using the DMA. The size distributions thus inferred were corrected for multiple charging of particles in the 100–500 nm size range. The size distribution thus obtained is shown in Figure 6b was obtained for 100 and 30 nm powders. For both the powders, we observed a broad size distribution with a nonspecific peak. This is attributed to the distribution in agglomerates produced after micron size agglomerates are broken down due to turbulence in the UV143H venturi pump. Particles were collected for SEM analysis downstream of the DMA with the voltage stepped between ~50 and 10000 V to collect all particles that contribute to the DMA-inferred size distribution. Figure 7 shows four typical particles of 100 nm powder. As can be seen, the particles, depending on the extent of de-agglomeration, can be of nominal size ~ 30 to ~500 nm. Similarly, in Figure 8, 30 nm powder particles are shown. It can be seen that the de-agglomeration process is important for generating truly nanoscale aerosol. Although, we have generated sub-0.4 μm particles, further work on designing a venturi pump that leads to higher turbulence level is warranted for efficient de-agglomeration.

As noted before, the sustained generation of aerosols in the sub-micro and micron size range is challenging due to high inter-particle adhesion energy. By employing ultrasonic energy in the form of standing waves, we observed particle liftoff from surfaces from long periods (~hours). The reflection of pressure waves from the surface of the powder bed are believed to cause the liftoff. In the future, detailed CFD analysis of the particle-pressure wave interaction will be carried to explain the particle lift off mechanism. From Figure 3, it can be seen that the ultrasonic lift off is very effective in sustaining aerosol generation. From prior investigations of particle re-entrainment from subsonic (Smedley, Phares, and Flagan 1999a, 2001) and supersonic jets (Smedley, Phares, and Flagan 1999b), turbulent channel flow (Goldasteh, Ahmadi, and Ferro 2013, 2015), it is known that fluid kinetic energy is an efficient mechanism for particle removal from surfaces. The rotating turntable design with grooves filled with dry powder used in this study is similar to the design of the TSI 3433 Small Scale Powder Disperser. However, the key difference is the use of an ultrasonic transducer to excite the powder and lift them off the surface to be sampled into a venturi pump for de-agglomeration. In principle, the ultrasonic dispersion described here can potentially enhance the aerosol generation using a TSI 3433 and increase the stability and concentration level achievable using dry powder dispersion. Schmoll et al. (2009) have reported that TiO_2 powders could not successfully aerosolized using the TSI SSPD and attribute it to the sticky or cohesive nature of TiO_2 particles. Here, we have shown that using the well-

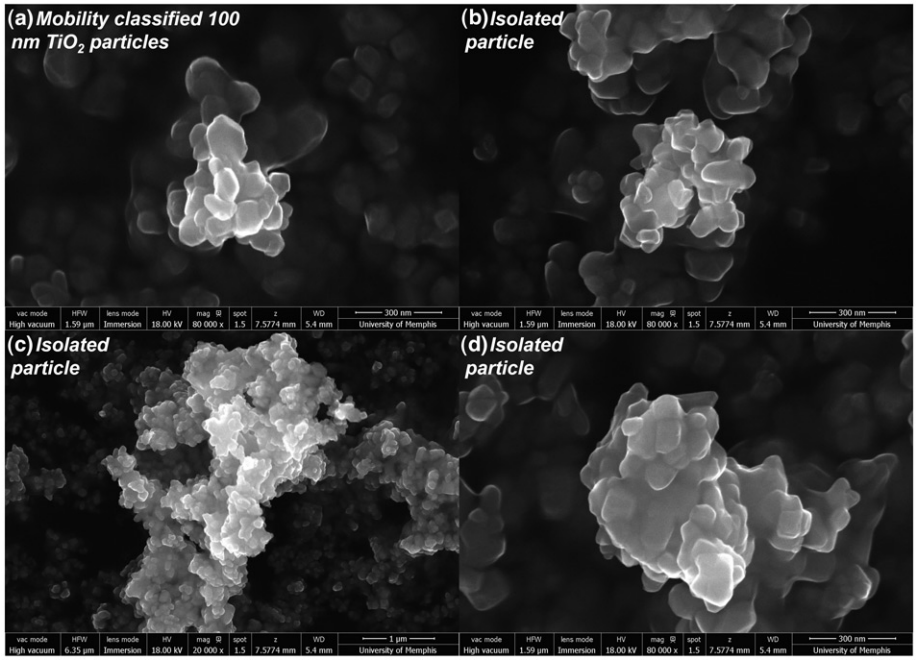


Figure 7. SEM images of particles collected downstream of the DMA for 100 nm TiO_2 aerosol particles when using UV 143H venturi pump.

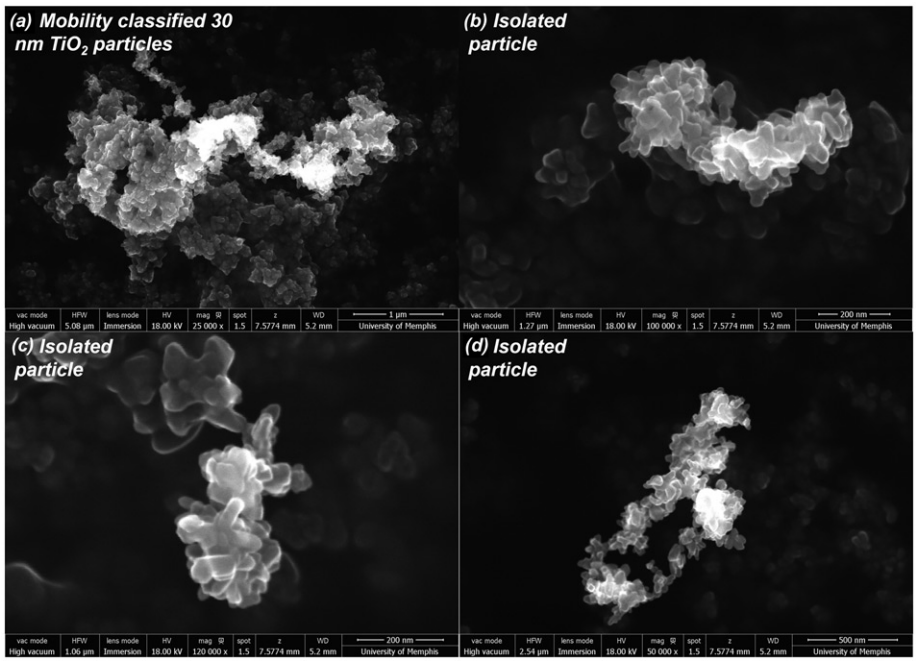


Figure 8. SEM images of particles collected downstream of the DMA for 30 nm TiO_2 aerosol particles when using UV 143H venturi pump.

established idea of venturi-based de-agglomeration aerosols at high concentration $\sim$5000 cm^{-3} can be generated for $\sim$ hour duration by steadily feeding particles by liftoff without moving parts. Aerosol generation in this study also strongly depends on the use of turbulent kinetic energy for de-agglomeration. Although, the two venturi pumps used in this study (TD260LSS and UV143H, both operated at 80 psig of inlet air pressure), lead to a broad size distribution in the sub-0.4 µm and 0.3–10 µm size range as inferred using electrical mobility analysis and the OPS, respectively. The marked difference between the size distributions in Figure 4 (obtained using TD260LSS) and Figure 6a (for the 500 nm, 100 nm and 30 nm TiO_2 powders obtained using UV143H) show the sensitivity of the de-agglomerated aerosol size distribution to the

turbulence intensity generated. From prior investigations, it is clear that the use of portable venturi pumps for de-agglomeration can be effective in producing nanoscale aerosol. The key advance in this experimental study is the proof of concept for a mechanism that can continuously act as a feeder of micro sized particles made of cohesive or sticky materials for a suitable, high efficiency de-agglomeration device. Further work is necessary to standardize the concentration output of the aerosol generation system and for fundamentally understanding the particle lift off mechanism using ultrasonic energy. In this study, we also ensured that the powder remains dry throughout the generation process. Although unexplored here, the effect of pre-milling the powder to break up large fragments is also expected to increase the dispersibility of the powders and needs to be investigated in the future. The width of the DMA-inferred size distribution of 100 and 30 nm powders shown in Figure 6b can be reduced by the use higher intensity jets for de-agglomeration and suitable dilution to reduce the possibility of coagulation of aerosol as they exit the venturi pump.

Conclusions

From this experimental study on aerosol generation from dry cohesive powders using 5 µm polyimide, 500, 100 and 30 nm TiO_2 powders, we draw the following conclusions:

1. A mechanism to lift powder particles from a bed is experimentally demonstrated. The use of ultrasonic energy, known for promoting coagulation and removal of pollutants from the air, is used here to create standing waves that liftoff particles to be sampled for aerosol generation. The dispersion of powders has been shown to continuously generate particles without any reduction in concentration for up to an hour in this study. Modeling of the flow field in the gap between the ultrasonic transducer and powder bed is necessary to understand the particle liftoff mechanism.
2. The concentration output is $\sim$4000–6000 cm^{-3} for different powders in this study. The design generated here is amenable to scale up and can be used to generate high concentration of micron-sized aerosols that is quite challenging if approached from bottom-up gas-to-particle conversion-based techniques. Further, dry powders are the favored form for handling raw materials in many industries.
3. The de-agglomeration of aerosol particles is sensitive to the turbulence intensity in the used portable venturi pumps. Further work is necessary to produce truly nanoscale aerosol based on past work on using simple vacuum generators for the same. The development of a continuous dry dispersion aerosol source has many applications in additive manufacturing and dosimetry studies as well.

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